

Unit 7 – Evaporation – McCabe and Smith

- Principle of Evaporation
- Types of evaporators- their construction and operation
- Natural circulation evaporators, short tube vertical or calandria type evaporators, basket type vertical evaporators, long tube vertical evaporators, forced circulation evaporators, falling film evaporators, climbing or rising film evaporators, agitated thin film evaporators, the plate evaporator.
- Single effect and multiple effect evaporators
- Performance of evaporators, capacity and economy of evaporators
- Overall heat transfer coefficient
- Effect of liquid head and boiling point elevation.
- Material and energy balances for single effect evaporator and the calculations on single effect evaporator. Multiple effect evaporators, Energy Balance.

Evaporation

- The objective of evaporation is to concentrate a solution consisting of a nonvolatile solute and a volatile solvent, majority of evaporations the solvent is water.
- Evaporation is conducted by vaporizing a portion of the solvent to produce a concentrated solution of thick liquor.
- Evaporation differs from drying in that the residue is a liquid-sometimes a highly viscous one-rather than a solid.
- Evaporation differs from distillation in that the vapor usually is a single component, and when the vapor is a mixture, no attempt is made in the evaporation step to separate the vapor into fractions.
- Evaporation differs from crystallization in that emphasis is placed on concentrating a solution rather than forming and building crystals.
- In certain situations, e.g., in the evaporation of brine to produce common salt, the line between evaporation and crystallization is far from sharp. Evaporation sometimes produces a slurry of crystals in a saturated mother liquor.

Evaporation

- Normally, in evaporation the thick liquor is the valuable product and the vapor is condensed and discarded.
- In one specific situation, however, the reverse is true.
- Mineral-bearing water often is evaporated to give a solid-free product for boiler feed, for special process requirements, or for human consumption.
- This technique is often called *water distillation*, but technically it is evaporation.
- Large-scale evaporation processes have been developed and used for recovering potable water from seawater. Here the condensed water is the desired product. Only a fraction of the total water in the feed is recovered, and the remainder is returned to the sea.

Factors affecting evaporator design

- **Liquid Characteristics:** The practical solution of an evaporation problem is profoundly affected by the character of the liquor to be concentrated.
- **Concentration:** Initially the thin liquor is sufficiently dilute to have density and viscosity of water. As concentration increases by evaporation, the density and viscosity increases as solution becomes saturated. Continued boiling of a saturated solution causes crystals to form which must be removed or the tubes clog.
- As solid content increases, the boiling point of the solution may also rise considerably, so that the boiling temperature of a concentrated solution may be much higher than that of water at the same pressure.
- **Foaming:** Some materials, especially organic substances, foam during vaporization. A stable foam accompanies the vapor out of the evaporator, causing heavy entrainment. In extreme cases the entire mass of liquid may boil over into the vapor outlet and be lost.

Factors affecting evaporator design

- **Temperature sensitivity:** Many fine chemicals, pharmaceutical products, and foods are damaged when heated to moderate temperatures for relatively short times. In concentrating such materials special techniques are needed to reduce both the temperature of the liquid and the time of heating.
- **Scale:** Some solutions deposit scale on the heating surface. The overall coefficient then steadily diminishes, until the evaporator must be shut down and the tubes cleaned. When the scale is hard and insoluble, the cleaning is difficult and **expensive**.
- **Materials of construction:** Whenever possible, evaporators are made of some kind of steel. Many solutions, however, attack ferrous metals or are contaminated by them. Special materials such as copper, nickel, stainless steel, aluminum, impervious graphite, and lead are then used.
- Since these materials are expensive, high heat-transfer rates become especially desirable to minimize the first cost of the equipment.

Factors affecting evaporator design

- Many other liquid characteristics must be considered during the design of an evaporator: specific heat, heat of concentration, freezing point, gas liberation on boiling, toxicity, explosion hazards, radioactivity, and necessity for sterile operation.

Notes Created by
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Single and Multiple-effect Operation

- When a single evaporator is used, the vapor from the boiling liquid is condensed and discarded.
- **This method is called *single-effect evaporation*, and although it is simple, it utilizes steam ineffectively.**
- To evaporate 1 kg of water from a solution calls for from 1 to 1.3 kg of steam.
- If the vapor from one evaporator is fed into the steam chest of a second evaporator and the vapor from the second is then sent to a condenser, the operation becomes double-effect.
- The heat in the original steam is reused in the second effect, and the evaporation achieved by a unit mass of steam fed to the first effect is approximately doubled.
- Additional effects can be added in the same manner.
- **The general method of increasing the evaporation per kilogram of steam by using a series of evaporators between the steam supply and the condenser is called *multiple-effect evaporation*.**

Multi-effect evaporator: <https://www.youtube.com/watch?v=yQQRzfnDgfQ>

Types of Evaporators

- Long-tube vertical evaporators:
 - a. Upward flow (climbing-film)
 - *b.* Downward flow (falling-film)
 - c. Forced circulation
- Agitated-film evaporators:

Once-through evaporators

- **In once-through operation the feed liquor passes through the tubes only once, releases the vapor, and leaves the unit as thick liquor. All the evaporation is accomplished in a single pass.**
- The ratio of evaporation to feed is limited in single-pass units; thus these evaporators are well adapted to multiple-effect operation, where the total amount of concentration can be spread over several effects.
- **Agitated-film evaporators are always operated once through; falling-film and climbing-film evaporators can also be operated in this way.**
- ***Once-through evaporators are especially useful for heat-sensitive materials. By operating under high vacuum, the temperature of the liquid can be kept low.***
- With a single rapid passage through the tubes the thick liquor is at the evaporation temperature but a short time and can be quickly cooled as soon as it leaves the evaporator.

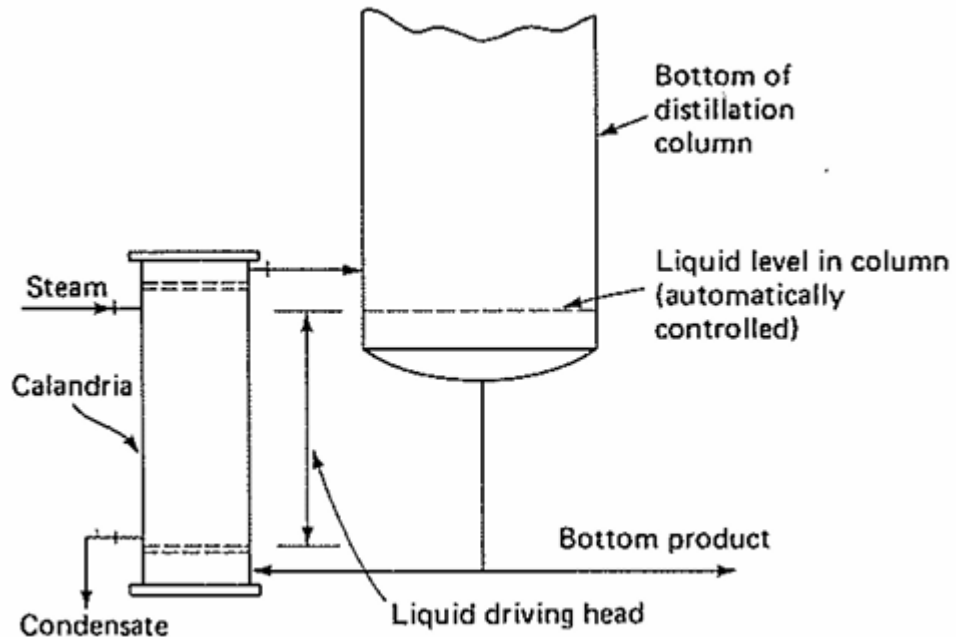
Circulation evaporators

- **In circulation evaporators a pool of liquid is held within the equipment. Incoming feed mixes with the liquid from the pool, and the mixture passes through the tubes.**
- **Unevaporated liquid discharged from the tubes returns to the pool, so that only part of the total evaporation occurs in one pass. All forced-circulation evaporators are operated in this way; climbing-film evaporators are usually circulation units.**
- The thick liquor from a circulation evaporator is withdrawn from the pool. All the liquor in the pool must therefore be at the maximum concentration.
- Since the liquid entering the tubes may contain several parts of thick liquor for each part of feed, its viscosity is high and the heat-transfer coefficient tends to be low.
- ***Circulation evaporators are not well suited to concentrating heat-sensitive liquids. With a reasonably good vacuum the temperature of the bulk of the liquid may be nondestructive, but the liquid is repeatedly exposed to contact with hot tubes.***

Circulation evaporators

- Some of the liquid, therefore, may be heated to an excessively high temperature.
- Although the average residence time of the liquid in the heating zone may be short, part of the liquid is retained in the evaporator for a considerable time. Prolonged heating of even a small part of a heat-sensitive material like a food can ruin the entire product.
- ***Circulation evaporators, however, can operate over a wide range of concentration between feed and thick liquor in a single unit, and are well adapted to single-effect evaporation.***
- ***They may operate either with natural circulation, with the flow through the tubes induced by density differences, or with forced circulation, with flow provided by a pump.***

Calendria evaporator

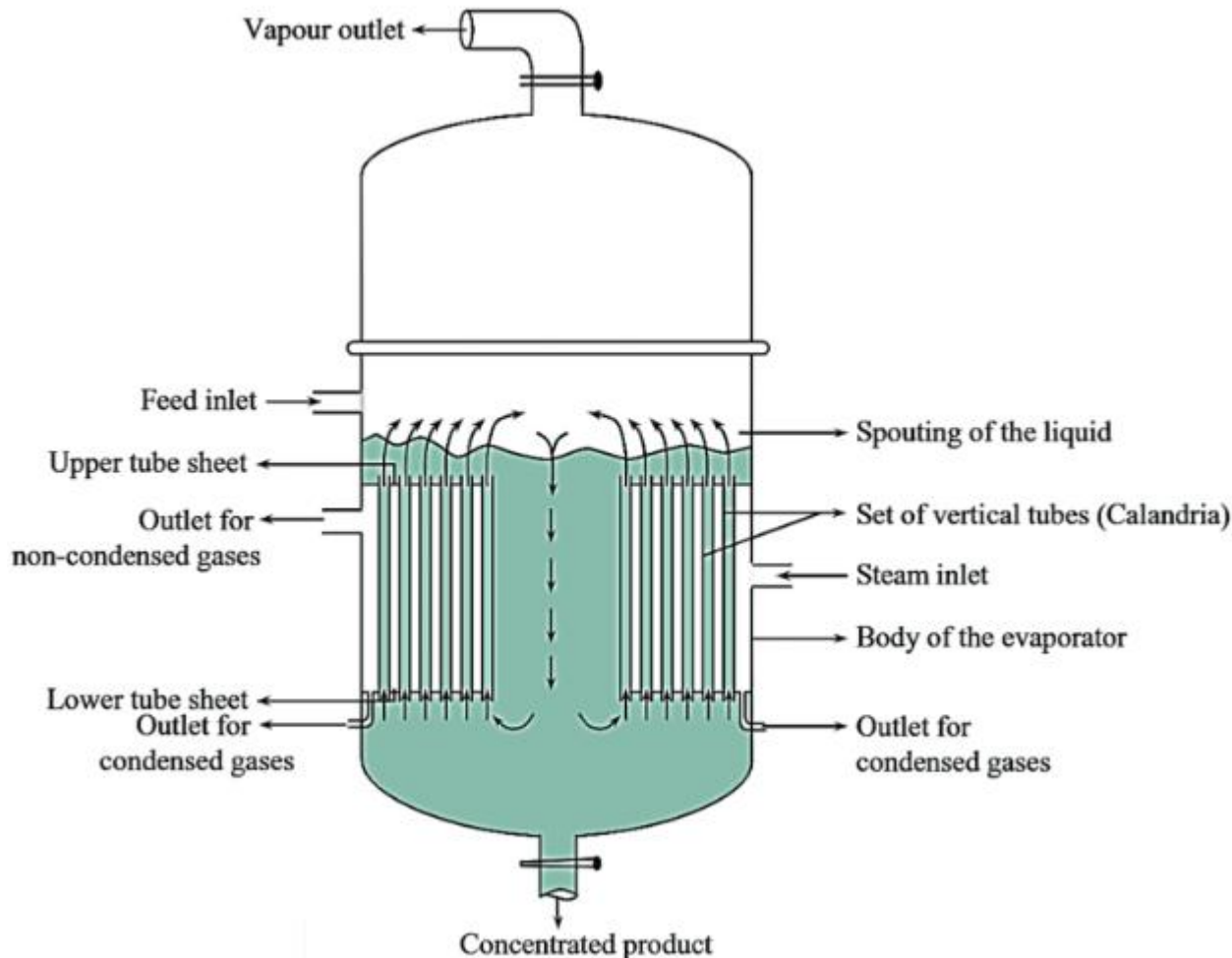


Typical overall coefficients in calandria boilers

Service	Overall coefficient U	
	$W/m^2 \cdot ^\circ C$	$Btu/ft^2 \cdot h \cdot ^\circ F$
Heavy organic chemicals	570–900	100–160
Light hydrocarbons	900–1250	160–220
Water, aqueous solutions	1250–2000	220–350

- In process plants liquids are boiled in kettle-type boilers containing a pool of boiling liquid or in **vertical-tube calandrias though which the liquid and vapor flow upward through the tubes.**
- Vertical shell-and-tube units, known as calandrias, natural-circulation or thermosiphon reboilers, are generally the most economical vaporizers for distillation and evaporation operations.
- **Liquid from an evaporator or distillation column enters the bottom of the unit and is partially vaporized in the heated tubes.**
- The reduction in density causes the vapor-liquid mixture to rise and draw in additional feed liquid.
- **Liquid and vapor leave the top of the tubes at high velocity; they are separated and the liquid is recycled.**

Calendria evaporator



- Liquid feed is introduced through the feed inlet and its level is maintained just above the tubes.
- Steam supplied through the steam inlet circulates in between the tubes and transfers its heat to the liquid within the tubes.
- Upon boiling, the liquid moves up through the tubes and then flows down the central compartment.
- Such movement helps to maintain the circulation of the liquid.
- The concentrated product formed is collected from the bottom of the evaporator.

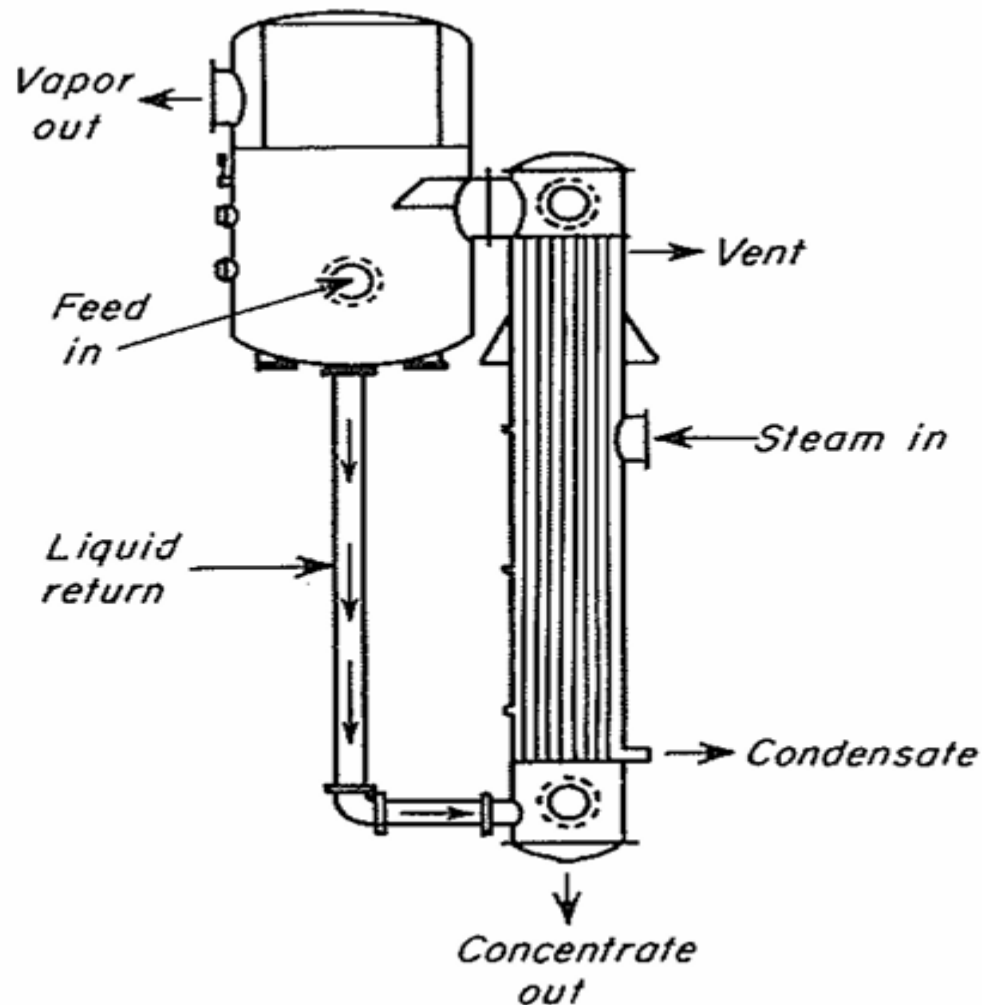
Calendria evaporator

- **The tubes are typically 25 mm in diameter**, sometimes as large as 50 mm.
- **They are 2.5 to 3.5 m long**, short tubes 1.2 to 1.8 m long are sometimes used in vacuum service.
- ***Heat-transfer rates depend on the properties of the vaporizing liquid, especially its reduced temperature T_c and on its tendency to foul the heat transfer surface.***
- They are relatively insensitive to changes in tube length or tube diameter.
- When the operating pressure is 1 atm abs or greater, the coefficients are also insensitive to changes in the liquid "driving head".
- This head is defined as the distance from the bottom tube sheet to the level of the liquid in the column.
- Under these pressure conditions the liquid level is typically maintained near the level of the top tube sheet to ensure that the heat-transfer surface is completely wetted and to establish reasonably high circulation rates.

Calendria evaporator

- When operation is under vacuum, however, the performance of the reboiler is sensitive to changes in the liquid driving head, especially in the distillation of multicomponent mixtures.
- The optimum liquid level for vacuum service is midway between the tube sheets, with about 50 percent of the liquid vaporized per pass.

Long-tube Evaporators With Upward Flow



- The essential parts are
 - (1) a tubular exchanger with steam in the shell and liquid to be concentrated in the tubes
 - (2) a separator or vapor space for removing entrained liquid from the vapor,
 - (3) when operated as a circulation unit, a return leg for the liquid from the separator to the bottom of the exchanger.
- Inlets are provided for feed liquid and steam, and outlets are provided for vapor, thick liquor, steam condensate, and noncondensable gases from the steam.

Long-tube Evaporators With Upward Flow

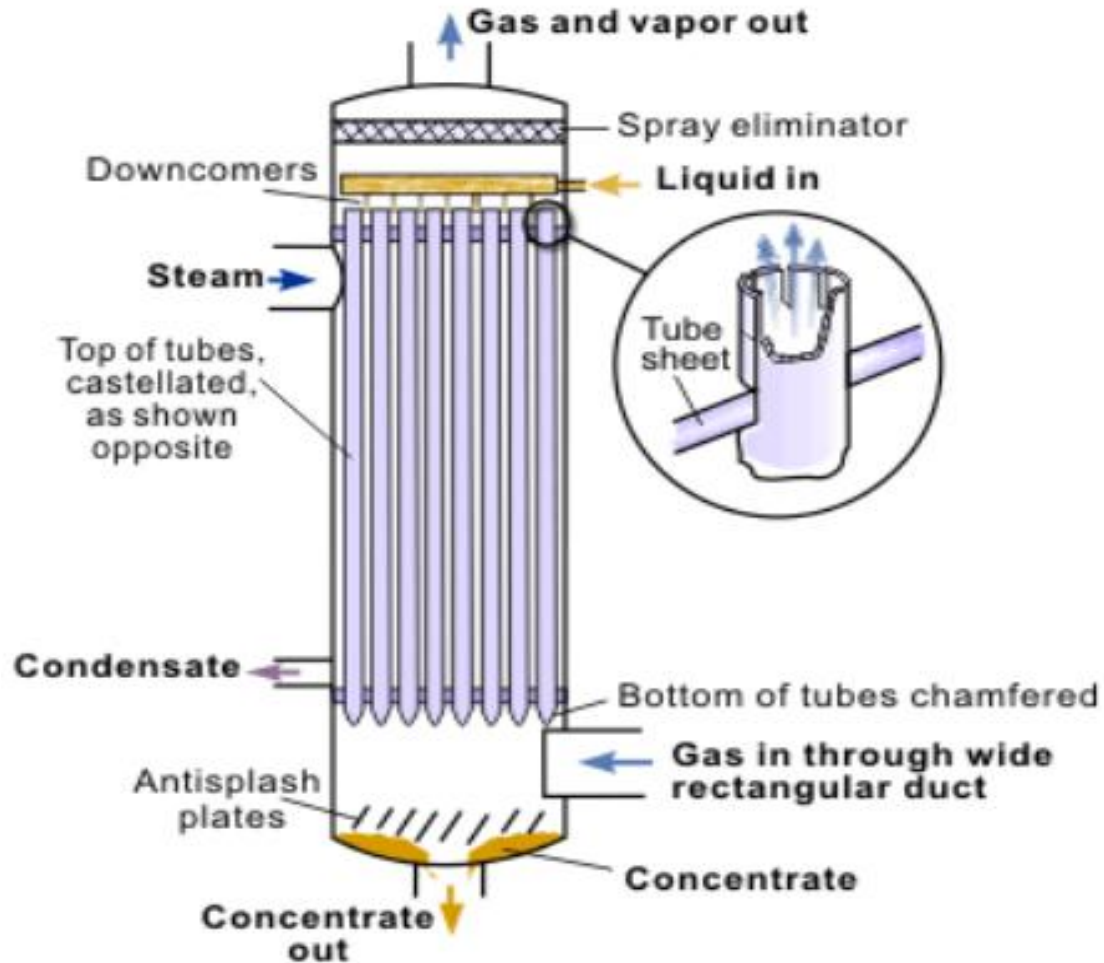
- *The tubular heater operates in exactly the same way as the natural circulation calandria.*
- *The tubes, however, are larger than in a calandria: they are typically 25 to 50 mm in diameter and 3 to 10 m long.*
- Dilute feed enters the system and mixes with the liquid draining from the separator.
- Concentrated liquor is withdrawn from the bottom of the heater; the remaining liquor is partially vaporized as it rises through the tubes.
- The mixture of liquid and vapor from the top of the tubes flows into the separator, where its velocity is greatly reduced.
- To aid in eliminating liquid droplets the vapor impinges on and then passes around sets of baffle plates before leaving the separator.
- The evaporator in Figure can be operated only as **a circulation unit.**

Long-tube Evaporators With Upward Flow

Application:

- Long-tube vertical evaporators are especially effective in concentrating liquids that tend to foam.
- Foam is broken when the high-velocity mixture of liquid and vapor impinges against the vapor-head baffle.

Falling-film Evaporators



- ***Concentration of highly heat-sensitive materials requires a minimum time of exposure to a heated surface.***
- ***This can be done in once-through falling-film evaporators, in which the liquid enters at the top, flows downstream inside the heated tubes as a film, and leaves from the bottom.***
- **The tubes are large, 50 to 250 mm in diameter.**
- Vapor evolved from the liquid is usually carried downward with the liquid and leaves from the bottom of the unit.
- In appearance these evaporators resemble long, vertical, tubular exchangers with a liquid-vapor separator at the bottom and a distributor for the liquid at the top.

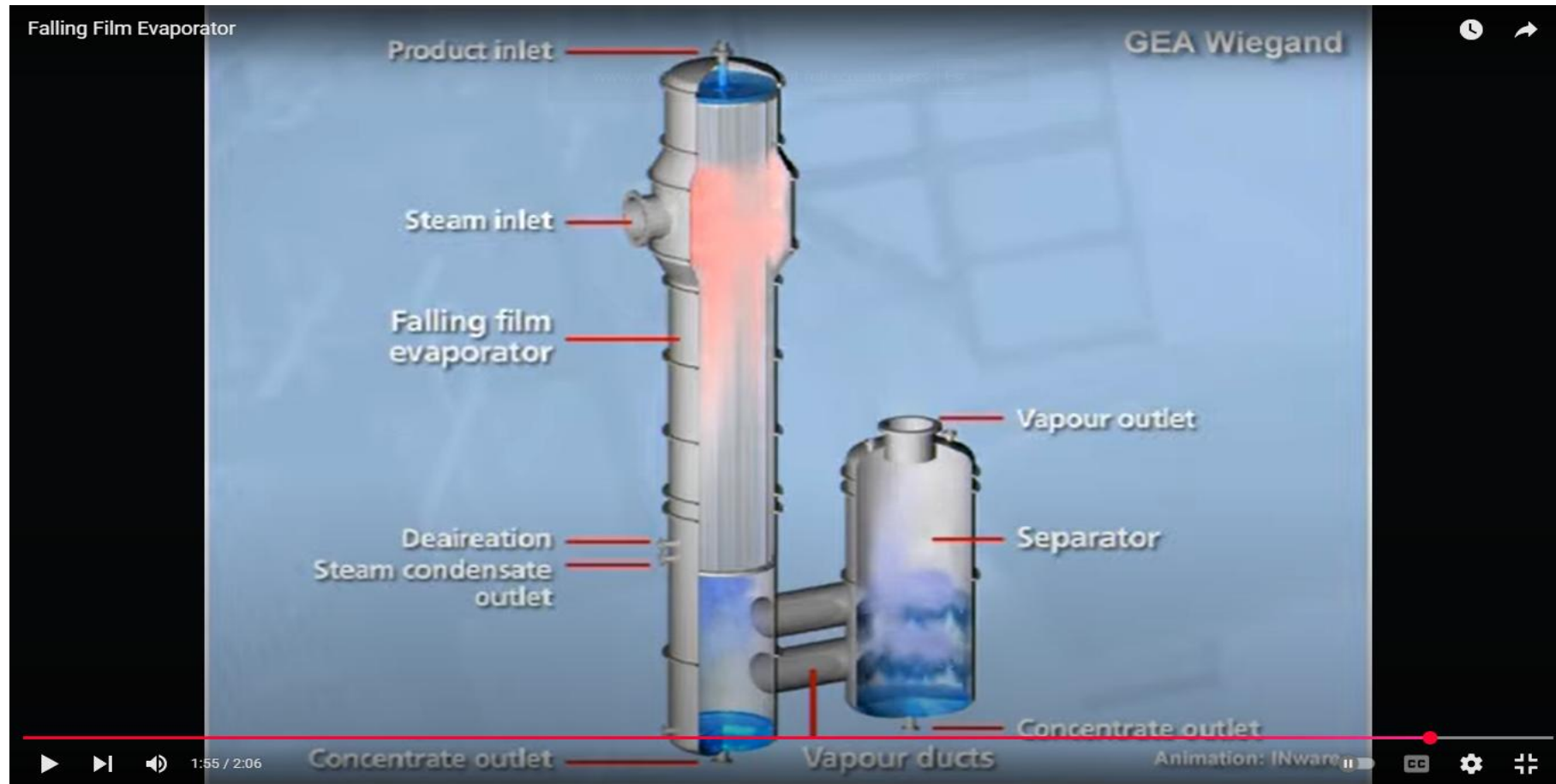
Falling-film Evaporators

- The chief **problem in a falling-film evaporator is that of distributing the liquid uniformly as a film inside the tubes.**
- This is done by a set of perforated metal plates above a carefully leveled tube sheet, by inserts in the tube ends to cause the liquid to flow evenly into each tube, or by "spider" distributors with radial arms from which the feed is sprayed at a steady rate on the inside surface of each tube.
- Still another way is to use an individual spray nozzle inside each tube.
- When recirculation is allowable without damaging the liquid, distribution of liquid to the tubes is facilitated by a moderate recycling of liquid to the tops of the tubes.
- This provides a larger volume of flow through the tubes than is possible in once-through operation.
- For good heat transfer the Reynolds number $4r/\mu$ of the falling film should be greater than 2000 at all points in the tube.

Falling-film Evaporators

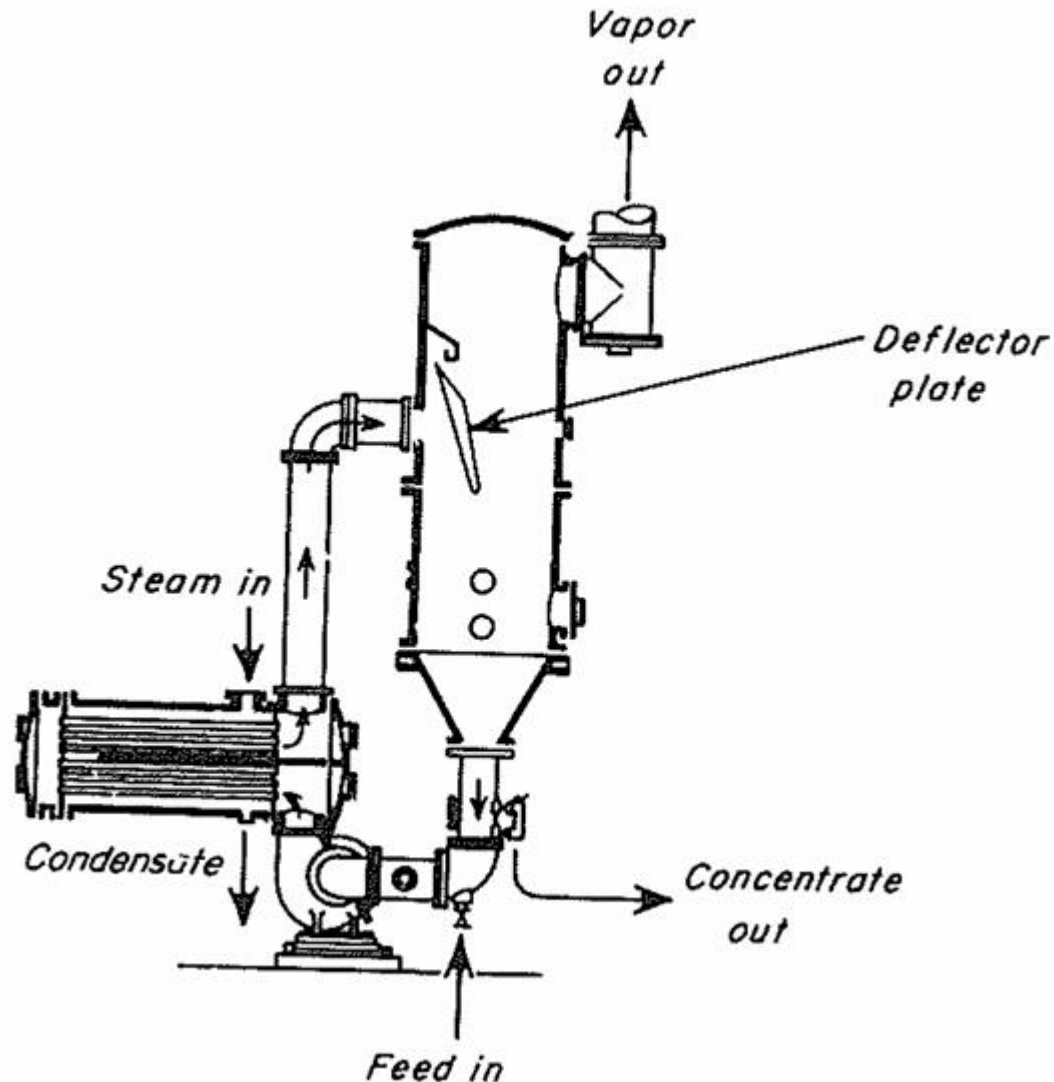
- During evaporation the amount of liquid is continuously reduced as it flows downward, and too great a reduction can lead to dry spots near the bottom of the tube.
- Thus the amount of concentration that can be done in a single pass is limited.
- **Application:**
- *Falling-film evaporators, with no recirculation and short residence times, handle sensitive products that can be concentrated in no other way.*
- *They are also well adapted to concentrating viscous liquids.*

Falling-film Evaporators



<https://www.youtube.com/watch?v=N7ilzKA5xh8>

Forced-circulation Evaporators



- In a natural-circulation evaporator the liquid enters the tubes at 0.3 to 1.2 m/s.
- The linear velocity increases greatly as vapor is formed in the tubes, so that in general the rates of heat transfer are satisfactory.
- **With viscous liquids, however, the overall coefficient in a natural circulation unit may be uneconomically low.**
- **Higher coefficients are obtained in forced-circulation evaporators.**
- ***Here a centrifugal pump forces liquid through the tubes at an entering velocity of 2 to 5.5 m/s.***

Forced-circulation Evaporators

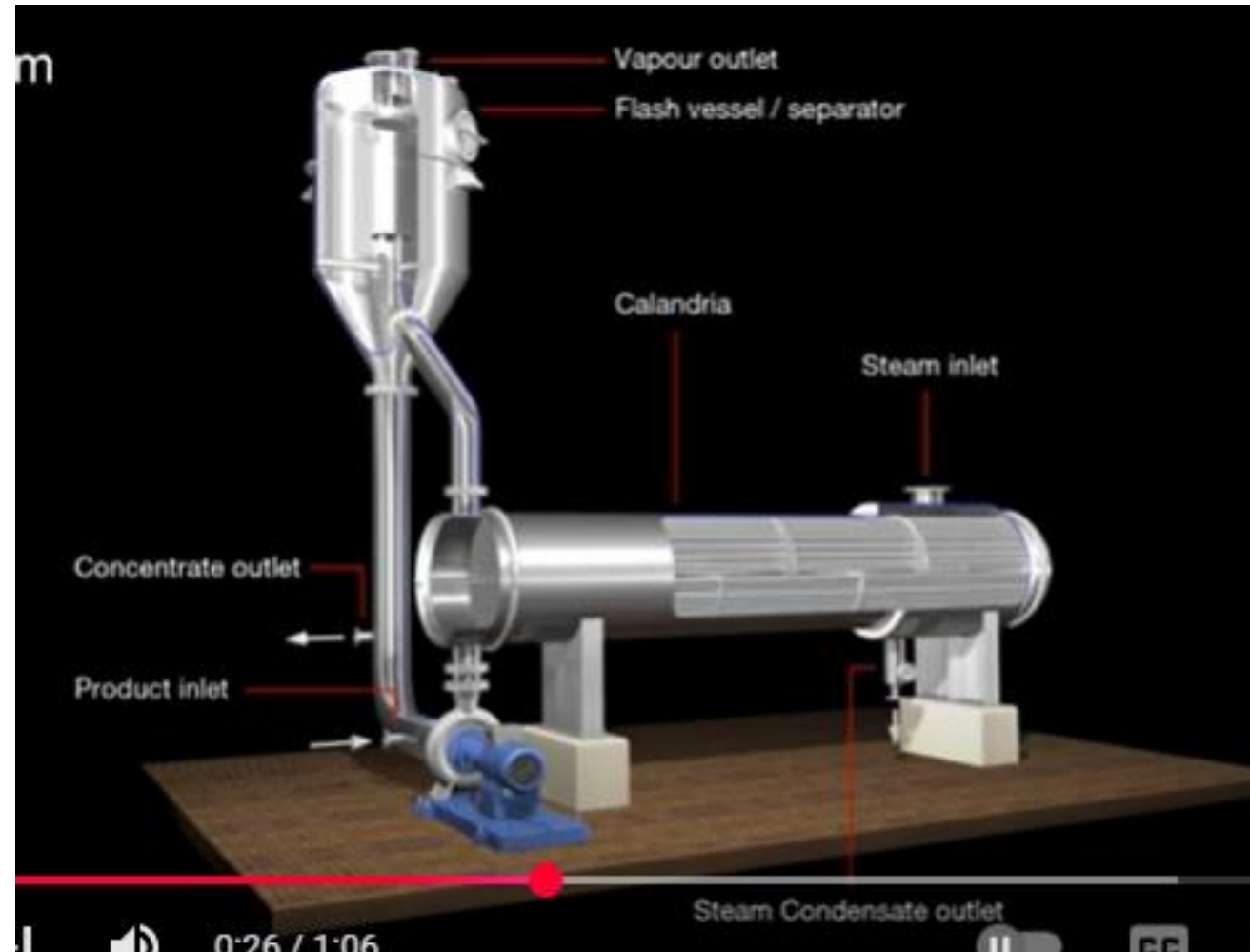
- The tubes are under sufficient static head to ensure that there is no boiling in the tubes.
- **The liquid becomes superheated as the static head is reduced during flow from the heater to the vapor space, and it flashes into a mixture of vapor and spray in the outlet line from the exchanger just before entering the body of the evaporator.**
- The mixture of liquid and vapor impinges on a deflector plate in the vapor space.
- Liquid returns to the pump inlet, where it meets incoming feed; vapor leaves the top of the evaporator body to a condenser or to the next effect.
- Part of the liquid leaving the separator is continuously withdrawn as concentrate.
- The exchanger can have horizontal tubes and is two-pass on both tube and shell sides.
- In others vertical single-pass exchangers are used.

Forced-circulation Evaporators

Applications:

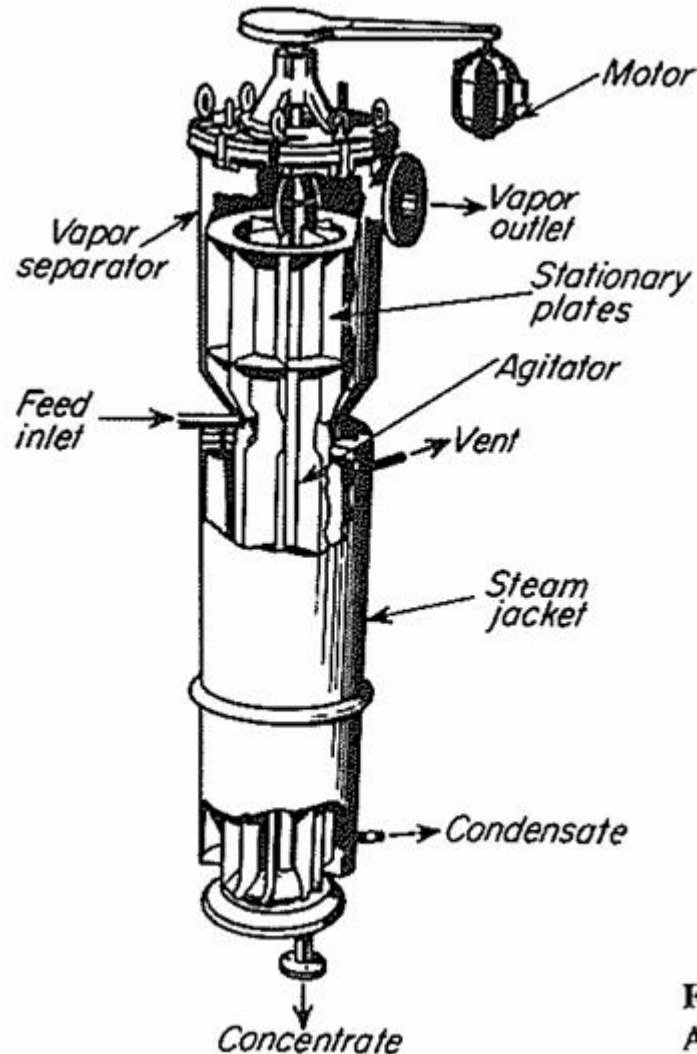
- **In both types the heat-transfer coefficients are high, especially with thin liquids, but the greatest improvement over natural-circulation evaporation is with viscous liquids.**
- With thin liquids the improvement with forced circulation does not warrant the added pumping costs over natural circulation, but with viscous material the added costs are justified, especially when expensive metals must be used.
- An example is caustic soda concentration, which must be done in nickel equipment.
- In multiple-effect evaporators producing a viscous final concentrate the first effects may be natural-circulation units and the later ones, handling viscous liquid, forced-circulation units.
- Because of the high velocities in a forced-circulation evaporator, the residence time of the liquid in the tubes is short-about 1 to 3 **so that moderately heat-sensitive liquids can be concentrated in them.**
- **They are also effective in evaporating salting liquors or those that tend to foam.**

Forced-circulation Evaporators



<https://www.youtube.com/watch?v=hCCLhwyodS4>

Agitated-film Evaporator



- The principal resistance to overall heat transfer from the steam to the boiling liquid in an evaporator is on the liquid side.
- One way of reducing this resistance, especially with viscous liquids, is by mechanical agitation of the liquid film.
- This is a modified falling-film evaporator with a single jacketed tube containing an internal agitator.

Agitated-film Evaporator

- **Feed enters at the top of the jacketed section and is spread out into a thin, highly turbulent film by the vertical blades of the agitator.**
- Concentrate leaves from the bottom of the jacketed section; vapor rises from the vaporizing zone into an unjacketed separator, which is somewhat larger in diameter than the evaporating tube.
- **In the separator the agitator blades throw entrained liquid outward against stationary vertical plates.**
- The droplets coalesce on these plates and return to the evaporating section.
- Liquid-free vapor escapes through outlets at the top of the unit.
- **The chief advantage of an agitated-film evaporator is its ability to give high rates of heat transfer with viscous liquids.**

Agitated-film Evaporator

- The product may have a viscosity as high as 1000 Pa.s at the evaporation temperature.
- **As in other evaporators, the overall coefficient falls as the viscosity rises, but in this design the decrease is slow.**
- ***With highly viscous materials the coefficient is appreciably greater than in forced-circulation evaporators and much greater than in natural circulation units.***
- ***The agitated-film evaporator is particularly effective with such viscous heat-sensitive products as gelatin, rubber latex, antibiotics, and fruit juices.***
- Its disadvantages are high cost; the internal moving parts, which may need considerable maintenance; and the small capacity of single units, which is far below that of multitubular evaporators.

Agitated-film Evaporator



<https://www.youtube.com/watch?v=lp1P8zEBMXI>

Performance of Tubular Evaporators

- The principal measures of the performance of a steam-heated tubular evaporator are the capacity and the economy.
- *Capacity* is defined as the number of kilograms of water vaporized per hour.
- $Capacity = \frac{\text{kilograms of water vaporized}}{\text{time (hour)}}$
- *Economy* is the number of kilograms vaporized per kilogram of steam fed to the unit.
- $Economy = \frac{\text{kilograms of water vaporized}}{\text{kilogram of steam fed to unit}}$
- In a single-effect evaporator the economy is nearly always less than 1, but in multiple-effect equipment it may be considerably greater.
- The steam consumption, in kilograms per hour, is also important.
- $Steam\ Consumption = \frac{\text{kilograms steam consumed}}{\text{time (hour)}} = \frac{Capacity}{Economy}$

Evaporator Capacity

- The rate of heat transfer q through the heating surface of an evaporator

$$q = UA \Delta T$$

1. If the **feed to the evaporator is at the boiling temperature** corresponding to the absolute pressure in the vapor space, **all the heat transferred through the heating surface is available for evaporation and the capacity is proportional to q .**
2. **If the feed is cold**, the heat required to heat it to its boiling point may be quite large and the **capacity for a given value of q is reduced accordingly**, as heat used to heat the feed is not available for evaporation.
3. Conversely, if the **feed is at a temperature above the boiling point in the vapor space**, a portion of the feed evaporates spontaneously by adiabatic equilibration with the vapor-space pressure and the **capacity is greater than that corresponding to q . This process is called flash evaporation.**

Evaporator Capacity

- **The actual temperature drop across the heating surface depends on the solution being evaporated, the difference in pressure between the steam chest and the vapor space above the boiling liquid, and the depth of liquid over the heating surface.**
- In some evaporators the **velocity of the liquid** in the tubes also influences the temperature drop because the frictional loss in the tubes increases the effective pressure of the liquid.
- When the solution has the characteristics of pure water, its boiling point can be read from steam tables if the pressure is known, as can the temperature of the condensing steam.
- **In actual evaporators, however, the boiling point of a solution is affected by two factors, boiling-point elevation and liquid head.**

Boiling-point Elevation and Duhring' S Rule

- The vapor pressure of aqueous solutions is less than that of water at the same temperature.
- Consequently, for a given pressure the boiling point of the solutions is higher than that of pure water.
- The increase in boiling point over that of water is known as the *boiling-point elevation* (BPE) of the solution.
- It is small for dilute solutions and for solutions of organic colloids but may be as large as 80 C (144 F) for concentrated solutions of inorganic salts.
- *The BPE must be subtracted from the temperature drop that is predicted from the steam tables.*

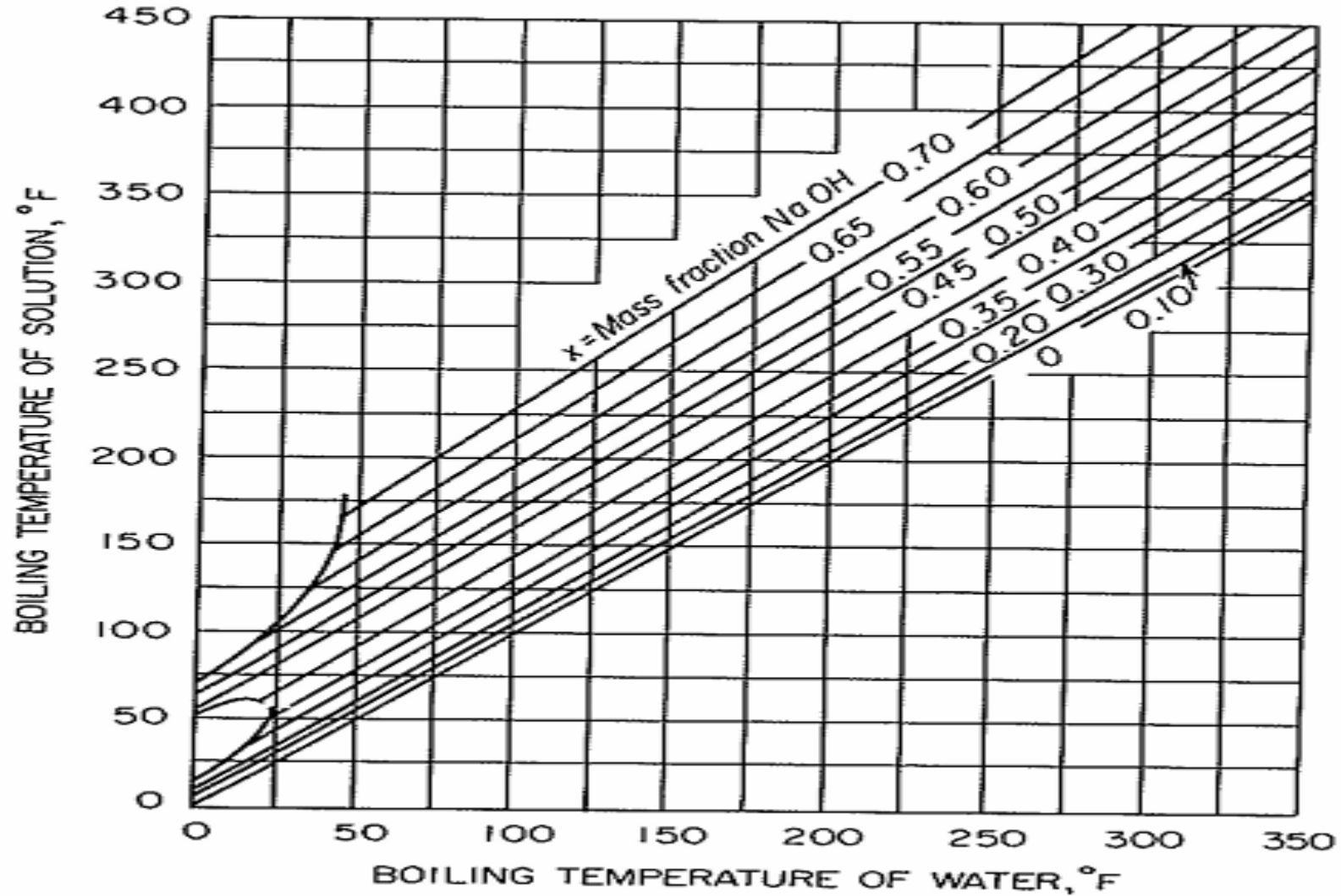
Boiling-point Elevation and Duhring' S Rule

- For strong solutions the BPE is best found from an empirical rule known as Duhring 's rule.
- **Duhring 's rule: states that the boiling point of a given solution is a linear function of the boiling point of pure water at the same pressure.**
- ***Thus if the boiling point of the solution is plotted against that of water at the same pressure, a straight line results.***
- ***Different lines are obtained for different concentrations.***
- Over wide ranges of pressure the rule is not exact, but over a moderate range the lines are very nearly straight, though not necessarily parallel.

Boiling-point Elevation and Duhring' S Rule

- Figure is a set of Duhring lines for solutions of sodium hydroxide in water.
- If the pressure over a 25 percent solution of sodium hydroxide is such that water boils at 180°F (82.2°C), by reading up from the x axis at 180°F to the line for 25 percent solution and then horizontally to the y axis it is found that the boiling point of the solution at this pressure is 200°F (93.3°C).
- The BPE for this solution at this pressure is therefore 20°F (11.1°C).

Boiling-point Elevation and Duhring' S Rule



Effect of Liquid Head and Friction on Temperature Drop

- If the depth of liquid in an evaporator is appreciable, the boiling point corresponding to the pressure in the vapor space is the boiling point of the surface layer of liquid only.
- ***A mass of liquid at a distance Z meters or feet below the surface is under a pressure of the vapor space plus a head of Z meters or feet of liquid and therefore has a higher boiling point.***
- In addition, when the velocity of the liquid is large, frictional loss in the tubes further increases the average pressure of the liquid.
- **In any actual evaporator, therefore, the average boiling point of the liquid in the tubes is higher than the boiling point corresponding to the pressure in the vapor space.**
- **This increase in boiling point lowers the average temperature drop between the steam and the liquid and reduces the capacity.**
- The amount of reduction cannot be estimated quantitatively with precision, but the qualitative effect of liquid head, especially with high liquor levels and high liquid velocities, should not be ignored.

Heat Transfer Coefficients

- **The overall coefficient, on the other hand, is strongly influenced by the design and method of operation of the evaporator.**
- The overall resistance to heat transfer between the steam and the boiling liquid is the sum of five individual resistances: the steam-film resistance; the two scale resistances, inside and outside the tubes; the tube-wall resistance; and the resistance from the boiling liquid.
- **Overall resistance = steam-film resistance + scale resistance inside tubes + scale resistance outside tubes + tube-wall resistance + resistance from boiling liquid.**
- The overall coefficient is the reciprocal of the overall resistance.
- **In most evaporators the fouling factor of the condensing steam and the resistance of the tube wall are very small, and they are usually neglected in evaporator calculations.**
- *In an agitated-film evaporator the tube wall is fairly thick, so that its resistance may be a significant part of the total.*

Steam-film coefficients

- **The steam-film coefficient is characteristically high, even when condensation is filmwise.**
- Promoters are sometimes added to the steam to give dropwise condensation and a still higher coefficient.
- Since the presence of non-condensable gas seriously reduces the steam-film coefficient, provision must be made to vent non-condensables from the steam chest and to prevent leakage of air inward when the steam is at a pressure below atmospheric.

Liquid-side coefficients

- **The liquid-side coefficient depends to a large extent on the velocity of the liquid over the heated surface.**
- In most evaporators, and especially those handling viscous materials, the resistance of the liquid side controls the overall rate of heat transfer to the boiling liquid.
- **In natural-circulation evaporators the liquid-side coefficient for dilute aqueous solutions is between 1500 and 3000 W/m²-°C.**
- **Forced circulation gives high liquid-side coefficients even though boiling inside the tubes is suppressed by the high static head.**
- The formation of scale on the tubes of an evaporator adds a thermal resistance equivalent to a fouling factor.

Overall Coefficients

- Because of the difficulty of measuring the individual film coefficients in an evaporator, experimental results are usually expressed in terms of overall coefficients.
- **These are based on the net temperature drop corrected for boiling-point elevation.**
- The overall coefficient is influenced by the same factors influencing individual coefficients; but if one resistance (say, that of the liquid film) is controlling, large changes in the other resistances have almost no effect on the overall coefficient.
- A small accumulation of scale reduces the coefficients to a small fraction of the clean-tube values.
- An agitated-film evaporator gives a seemingly low coefficient with a liquid having a viscosity of 100 P, but this coefficient is much larger than would be obtained in any other type of evaporator which could handle such a viscous material at all.

Overall Coefficients

Typical overall coefficients in evaporators

Type	Overall coefficient U	
	$W/m^2\text{-}^\circ C$	$Btu/ft^2\text{-}h\text{-}^\circ F$
Long-tube vertical evaporators		
Natural circulation	1000–2500	200–500
Forced circulation	2000–5000	400–1000
Agitated-film evaporator, newtonian liquid, viscosity		
1 cP	2000	400
1 P	1500	300
100 P	600	120

Evaporator Economy

- **The chief factor influencing the economy of an evaporator system is the number of effects.**
- By proper design the enthalpy of vaporization of the steam to the first effect can be used one or more times, depending on the number of effects.
- The economy also is influenced by the temperature of the feed.
- If the temperature is below the boiling point in the first effect, the heating load uses a part of the enthalpy of vaporization of the steam and only a fraction is left for evaporation.
- If the feed is at a temperature above the boiling point, the accompanying flash contributes some evaporation over and above that generated by the enthalpy of vaporization in the steam.
- **Quantitatively, evaporator economy is entirely a matter of enthalpy balances.**

Enthalpy Balances for Single-effect Evaporator

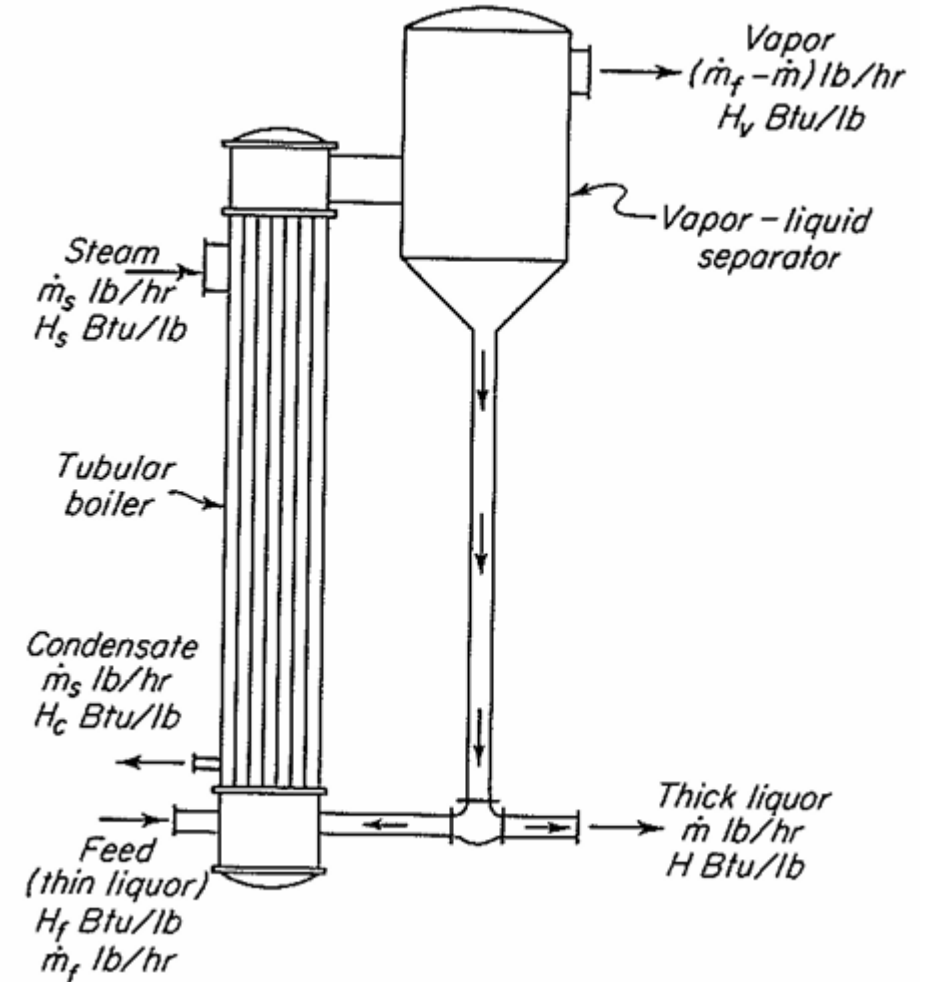
- In a single-effect evaporator the latent heat of condensation of the steam is transferred through a heating surface to vaporize water from a boiling solution.
- Two enthalpy balances are needed, one for the steam and one for the vapor or liquid side.

For a vertical-tube single-effect evaporator:

- Rate of steam flow and of condensate: $\dot{m}_s$
- Rate of thin liquor, or feed: $\dot{m}_f$
- Rate of the thick liquor: $\dot{m}$
- Rate of vapor flow to the condenser: $\dot{m}_f - \dot{m}$

(assuming that no solids precipitate from the liquor)

- Condensing temperature of the steam: T_s
- Boiling temperature of the liquid in the evaporator: T
- Temperature of the feed: T_f



Enthalpy Balances for Single-effect Evaporator

Assumptions:

- It is assumed that there is no leakage or entrainment, that the flow of noncondensables is negligible, and that heat losses from the evaporator need not be considered.
- The steam entering the steam chest may be superheated, and the condensate usually leaves the steam chest somewhat subcooled below its boiling point.
- Both the superheat and the subcooling of the condensate are small, however, and it is acceptable to neglect them in making an enthalpy balance.
- The small errors made in neglecting them are approximately compensated by neglecting heat losses from the steam chest.

Enthalpy Balances for Single-effect Evaporator

- Under these assumptions the difference between the enthalpy of the steam and that of the condensate is: λ_s
- Where λ_s is the latent heat of condensation of the steam.
- The enthalpy balance for the steam side is

$$q_s = \dot{m}_s(H_s - H_c) = \dot{m}_s\lambda_s$$

where q_s = rate of heat transfer through heating surface from steam

H_v = specific enthalpy of steam

H_c = specific enthalpy of condensate

λ_s = latent heat of condensation of steam

$\dot{m}_s$ = rate of flow of steam

Enthalpy Balances for Single-effect Evaporator

- The enthalpy balance for the liquor side is

$$q = (\dot{m}_f - \dot{m})H_v - \dot{m}_f H_f + \dot{m}H$$

where q = rate of heat transfer from heating surface to liquid

H_v = specific enthalpy of vapor

H_f = specific enthalpy of thin liquor

H = specific enthalpy of thick liquor

Enthalpy Balances for Single-effect Evaporator

- In the absence of heat losses, the heat transferred from the steam to the tubes equals that transferred from the tubes to the liquor
- and $q_s = q$.
- Thus, by combining Eqs.

$$q = \dot{m}_s \lambda_s = (\dot{m}_f - \dot{m})H_v - \dot{m}_f H_f + \dot{m}H$$

Example

- A single-effect evaporator is to concentrate 20,000 lb_fh (9070 kg_fh) of a 20 percent solution of sodium hydroxide to 50 percent solids.
- The gauge pressure of the steam is to be 20 lb₁/in.² (1.37 atm); the absolute pressure in the vapor space is to be 100 mm Hg (1.93 lb_f/in.²).
- The overall coefficient is estimated to be 250 Btu/ft²-h-°F (1400 W/m²-C).
- The feed temperature is 100°F (37.8°C).
- Calculate the amount of steam consumed, the economy, and the heating surface required.

Solution

The amount of water evaporated is found from a material balance. The feed contains $\frac{80}{20} = 4$ lb of water per pound of solid; the thick liquor contains $\frac{50}{50} = 1$ lb of water per pound of solid. The quantity evaporated is $4 - 1 = 3$ lb of water per pound of solid, or

$$3 \times 20,000 \times 0.20 = 12,000 \text{ lb/h}$$

The flow rate of thick liquor $\dot{m}$ is

$$20,000 - 12,000 = 8000 \text{ lb/h (3630 kg/h)}$$

Steam consumption Since with strong solutions of sodium hydroxide the heat of dilution is not negligible, the rate of heat transfer is found from Eq. (16.4) and Fig. 16.8. The vaporization temperature of the 50 percent solution at a pressure of 100 mm Hg is found as follows.

$$\text{Boiling point of water at 100 mm Hg} = 124^{\circ}\text{F} \quad (\text{Appendix 7})$$

$$\text{Boiling point of solution} = 197^{\circ}\text{F} \quad (\text{Fig. 16.4})$$

$$\text{Boiling-point elevation} = 197 - 124 = 73^{\circ}\text{F}$$

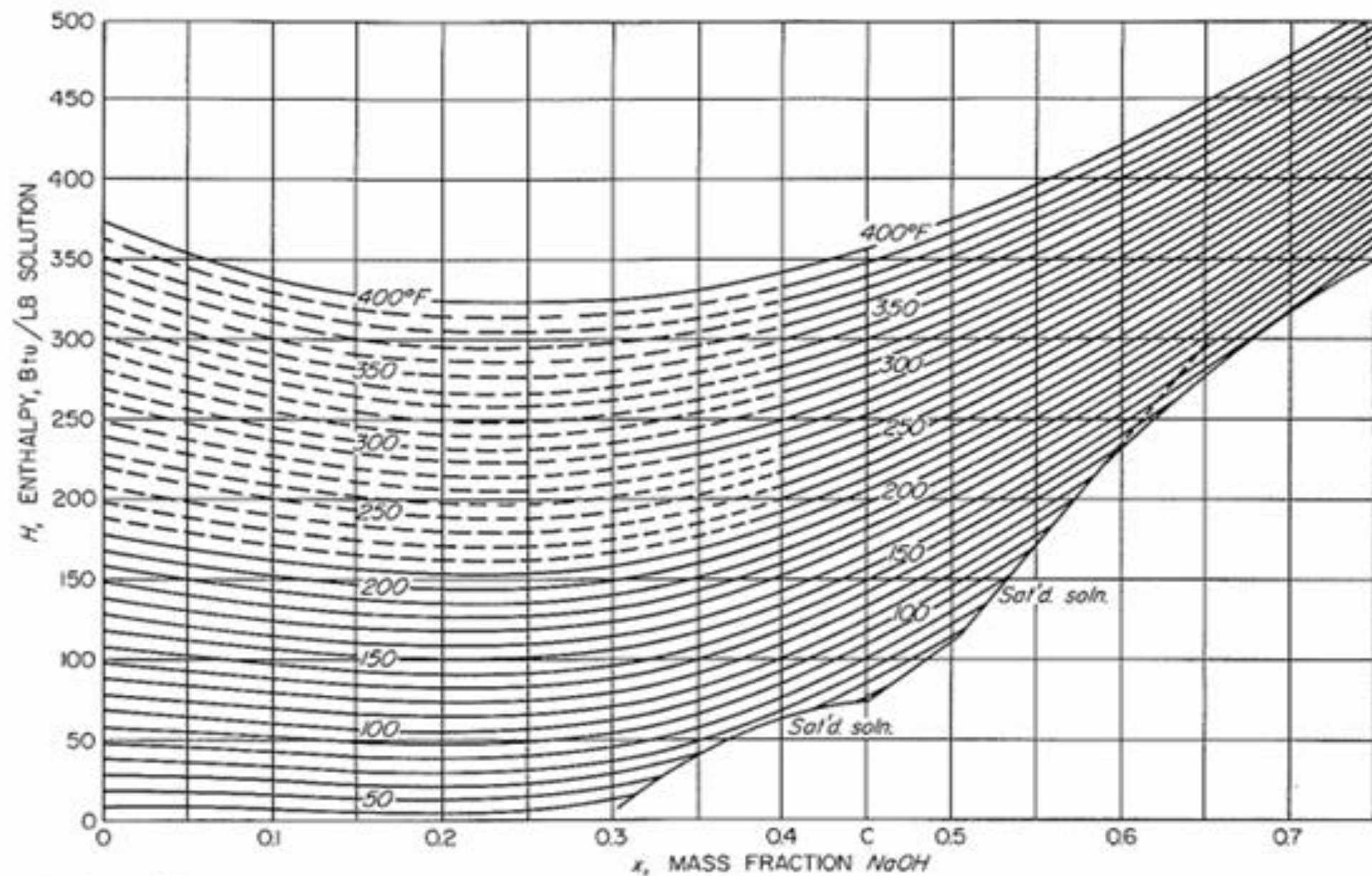


FIGURE 16.8

Enthalpy-concentration diagram, system sodium hydroxide—water. Dashed lines represent extrapolated data. (After McCabe⁶.)

The enthalpies of the feed and thick liquor are found from Fig. 16.8:

$$\text{Feed, 20\% solids, } 100^{\circ}\text{F:} \quad H_f = 55 \text{ Btu/lb}$$

$$\text{Thick liquor, 50\% solids, } 197^{\circ}\text{F:} \quad H = 221 \text{ Btu/lb}$$

The enthalpy of the vapor leaving the evaporator is found from steam tables. The enthalpy of superheated water vapor at 197°F and $1.93 \text{ lb}_f/\text{in.}^2$ is 1149 Btu/lb; this is the H_v of Eq. (16.4).

The heat of vaporization of steam λ_s at a gauge pressure of $20 \text{ lb}_f/\text{in.}^2$ is, from Appendix 7, 939 Btu/lb.

The rate of heat transfer and the steam consumption can now be found from Eq. (16.4):

$$q = (20,000 - 8000)(1149) + 8000 \times 221 - 20,000 \times 55 = 14,456,000 \text{ Btu/h}$$

$$\dot{m}_s = \frac{14,456,000}{939} = 15,400 \text{ lb/h (6990 kg/h)}$$

Economy The economy is $12,000/15,400 = 0.78$.

Heating surface The condensation temperature of the steam is 259°F . The heating area required is

$$A = \frac{14,456,000}{250(259 - 197)} = 930 \text{ ft}^2 (86.4 \text{ m}^2)$$

Multiple-effect Evaporators

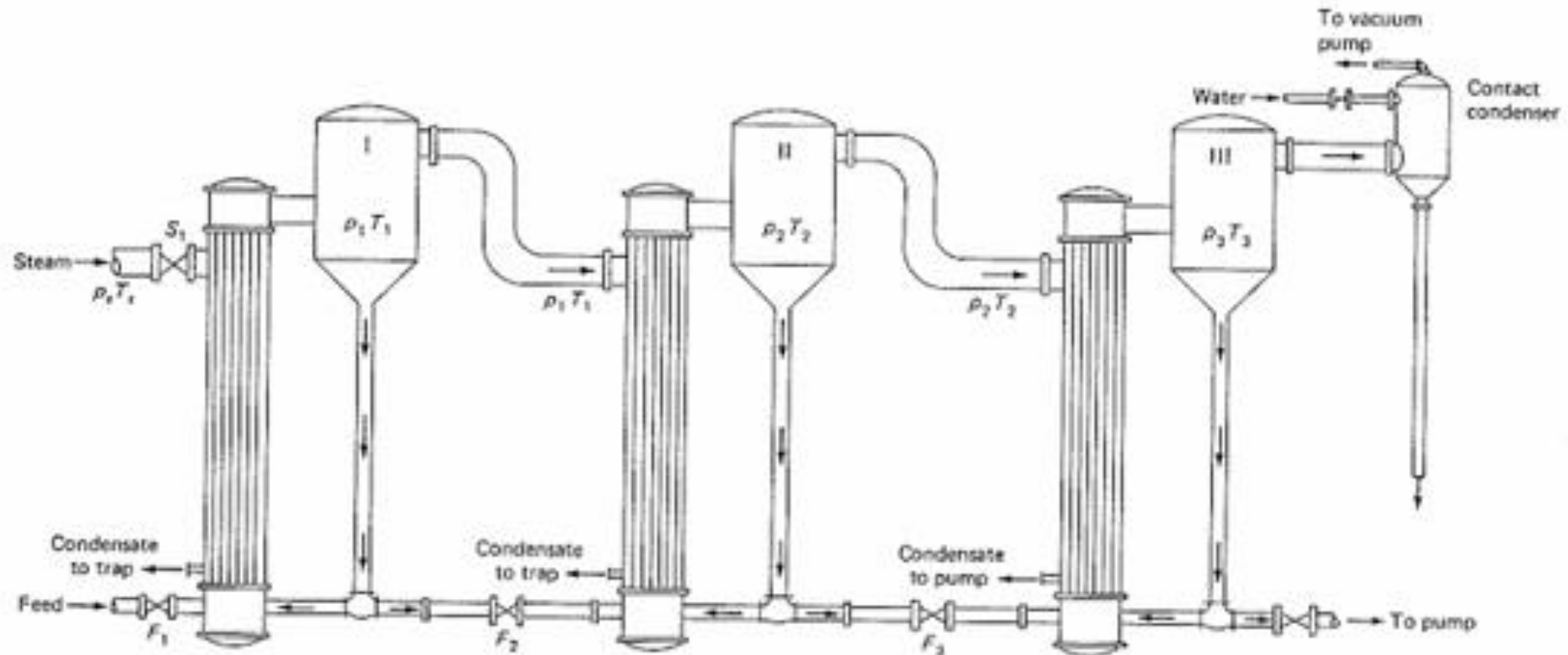


FIGURE 16.9

Triple-effect evaporator: I, II, III, first, second, and third effects; F_1, F_2, F_3 , feed or liquor control valves; S_1 , steam valve; p_s, p_1, p_2, p_3 , pressures; T_s, T_1, T_2, T_3 , temperatures.

Multiple-effect Evaporators

- Figure shows three long-tube natural circulation evaporators connected together to form a triple-effect system.
- Dilute feed enters the first effect, where it is partly concentrated; it flows to the second effect for additional concentration and then to the third effect for final concentration. Thick liquor is pumped out of the third effect.
- Connections are made so that the vapor from one effect serves as the heating medium for the next.
- A condenser and air ejector establish a vacuum in the third effect in the series and withdraw noncondensables from the system.
- The first effect of a multiple-effect evaporator is the effect to which the raw steam is fed and in which the pressure in the vapor space is the highest.
- The last effect is that in which the vapor-space pressure is a minimum.

Multiple-effect Evaporators

- In this manner the pressure difference between the steam and the condenser is spread across two or more effects in the multiple effect system.
- The pressure in each effect is lower than that in the effect from which it receives steam and above that of the effect to which it supplies vapor.
- Each effect, in itself, acts as a single-effect evaporator, and each has a temperature drop across its heating surface corresponding to the pressure drop in that effect.
- Every statement that has so far been made about a single-effect evaporator applies to each effect of a multiple-effect system.
- Arranging a series of evaporator bodies into a multiple-effect system is a matter of interconnecting piping, and not of the structure of the individual units.
- The numbering of the effects is independent of the order liquor is fed to them-they are always numbered in the direction of decreasing pressure.

Multiple-effect Evaporators

- In steady operation the flow rates and evaporation rates are such that neither solvent nor solute accumulates or depletes in any of the effects.
- The heating surface in the first effect will transmit per hour an amount of heat given by the equation

$$q_1 = A_1 U_1 \Delta T_1$$

- If the part of this heat that goes to heat the feed to the boiling point is neglected, all of q_1 heat must appear as latent heat in the vapor that leaves the first effect.
- The temperature of the condensate leaving the second effect is very near the temperature T_1 of the vapors from the boiling liquid in the first effect.
- Therefore, in steady operation practically all of the heat that was expended in creating vapor in the first effect must be given up when this same vapor condenses in the second effect.

Multiple-effect Evaporators

- The heat transmitted in the second effect, however, is given by the equation

$$q_2 = A_2 U_2 \Delta T_2$$

- q_1 and q_2 are nearly equal, and therefore

$$A_1 U_1 \Delta T_1 = A_2 U_2 \Delta T_2$$

- When extended to 3 evaporators

$$A_1 U_1 \Delta T_1 = A_2 U_2 \Delta T_2 = A_3 U_3 \Delta T_3$$

Multiple-effect Evaporators

- In ordinary practice the heating areas in all the effects of a multiple effect evaporator are equal.
- This is to obtain economy of construction.
- Therefore, it follows that, since $q_1 = q_2 = q_3 = q$

$$U_1 \Delta T_1 = U_2 \Delta T_2 = U_3 \Delta T_3 = \frac{q}{A}$$

- From this it follows that the temperature drops in a multiple-effect evaporator are approximately inversely proportional to the heat-transfer coefficients.

Multiple-effect Evaporators

Example 16.2. A triple-effect evaporator is concentrating a liquid that has no appreciable elevation in boiling point. The temperature of the steam to the first effect is 108°C, the boiling point of the solution in the last effect is 52°C. The overall heat-transfer coefficients, in W/m²-°C, are 2500 in the first effect, 2000 in the second effect, and 1000 in the third effect. At what temperatures will the liquid boil in the first and second effects?

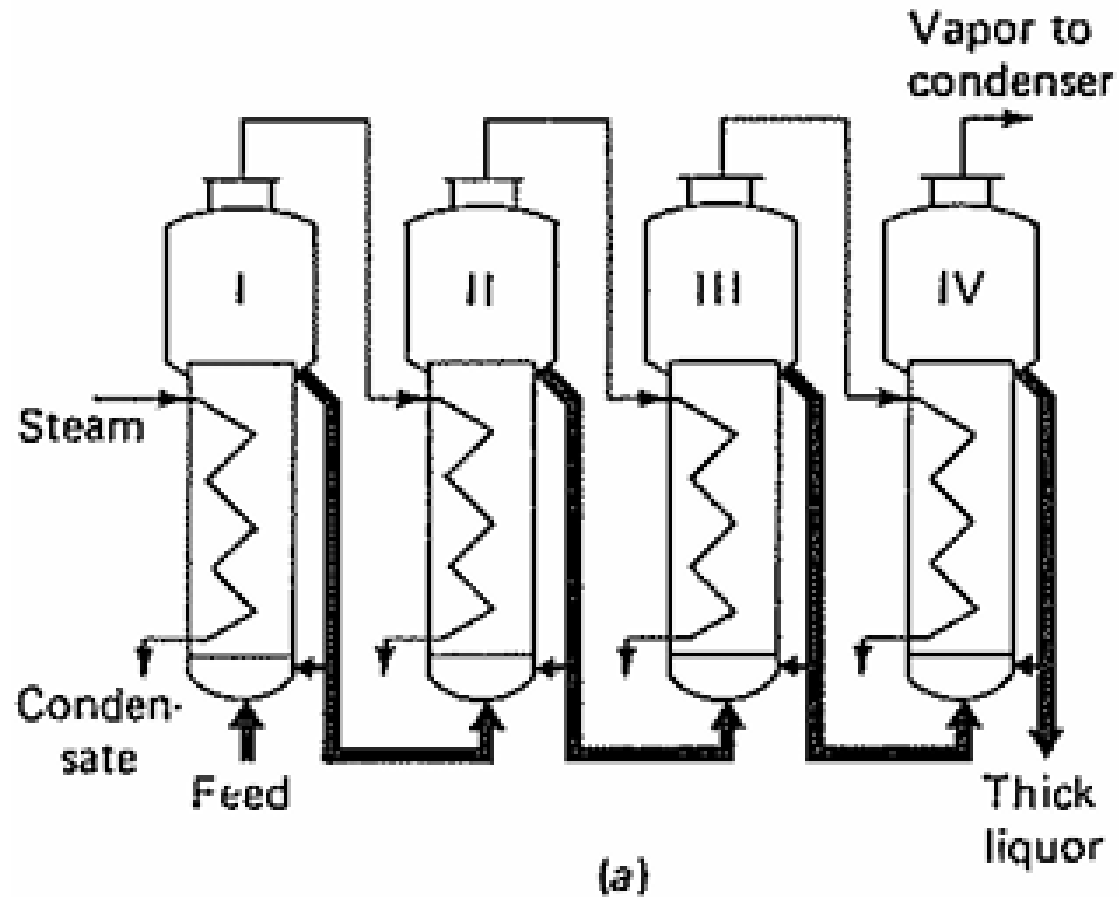
Solution

The total temperature drop is 108 – 52 = 56°C. As shown by Eq. (16.13), the temperature drops in the several effects will be approximately inversely proportional to the coefficients. Thus, for example,

$$\Delta T_1 = \frac{\frac{1}{2500}}{\frac{1}{2500} + \frac{1}{2000} + \frac{1}{1000}} 56 = 11.8^\circ\text{C}$$

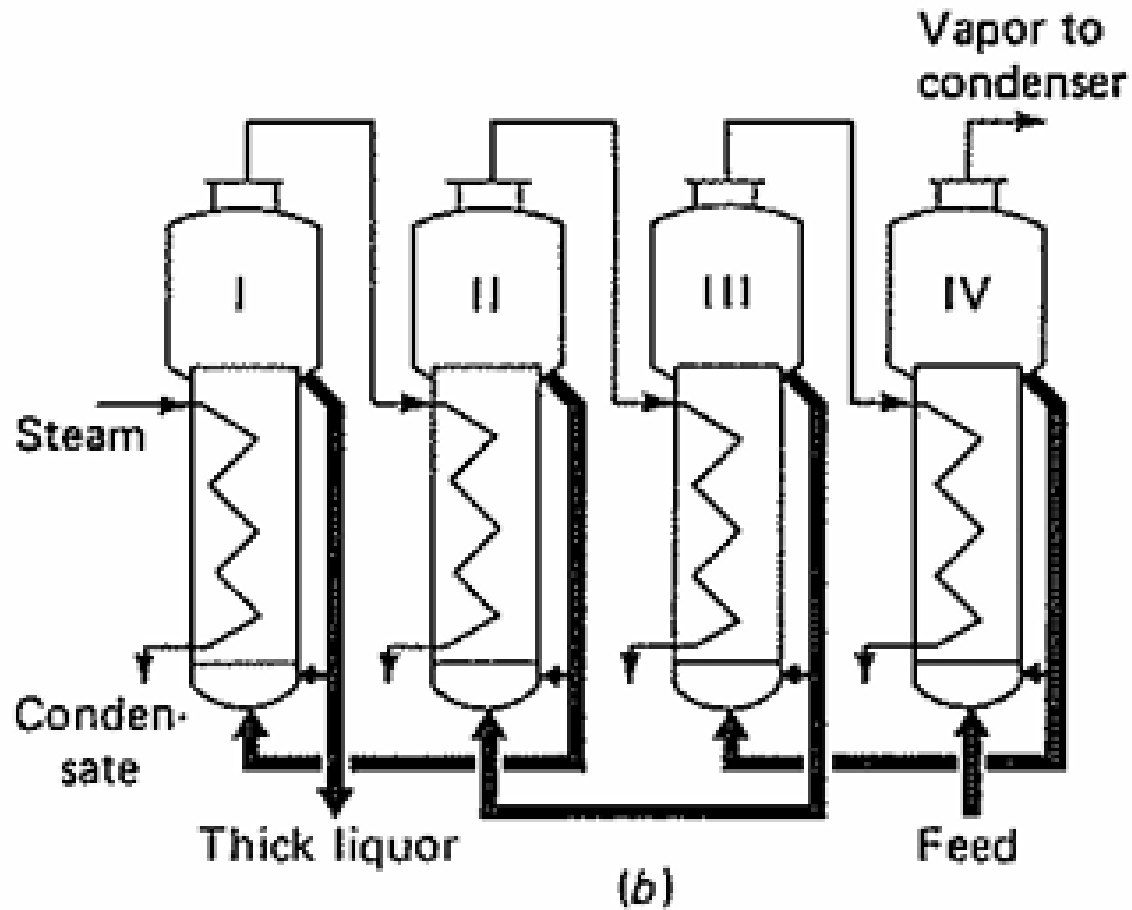
In the same manner $\Delta T_2 = 14.7^\circ\text{C}$ and $\Delta T_3 = 29.5^\circ\text{C}$. Consequently the boiling point in the first effect will be 108 – 11.8 = 96.2°C, and that in the second effect, 81.5°C.

Methods of Feeding – Forward Feed



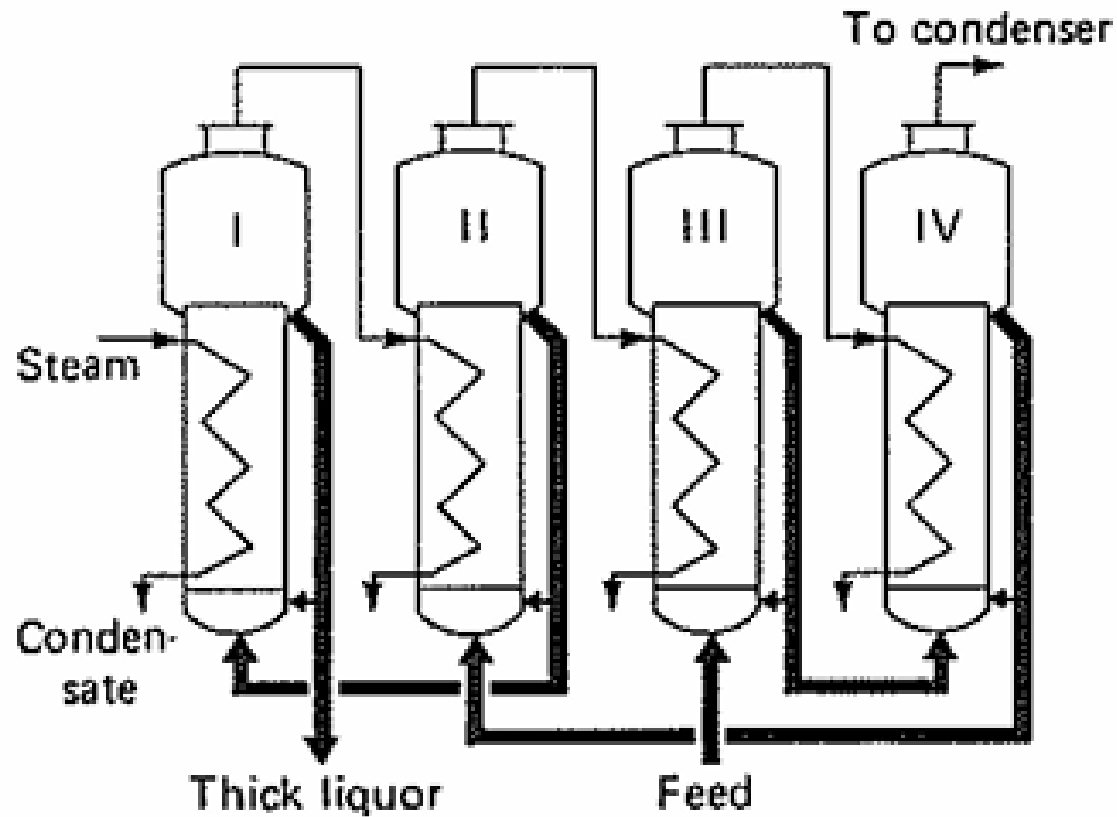
- The usual method of feeding a multiple-effect evaporator is to pump the thin liquid into the first effect and send it in turn through the other effects, is called *forward feed*.
- The concentration of the liquid increases from the first effect to the last. This pattern of liquid flow is the simplest.
- It requires a pump for feeding dilute solution to the first effect, since this effect is often at about atmospheric pressure, and a pump to remove thick liquor from the last effect.
- The transfer from effect to effect, however, can be done without pumps, since the flow is in the direction of decreasing pressure, and control valves in the transfer line are all that is required.

Methods of Feeding – Backward Feed



- Another common method is *backward feed*, in which dilute liquid is fed to the last effect and then pumped through the successive effects to the first.
- This method requires a pump between each pair of effects in addition to the thick-liquor pump, since the flow is from low pressure to high pressure.
- Backward feed often gives a higher capacity than forward feed when the thick liquor is viscous, but it may give a lower economy than forward feed when the feed liquor is cold.

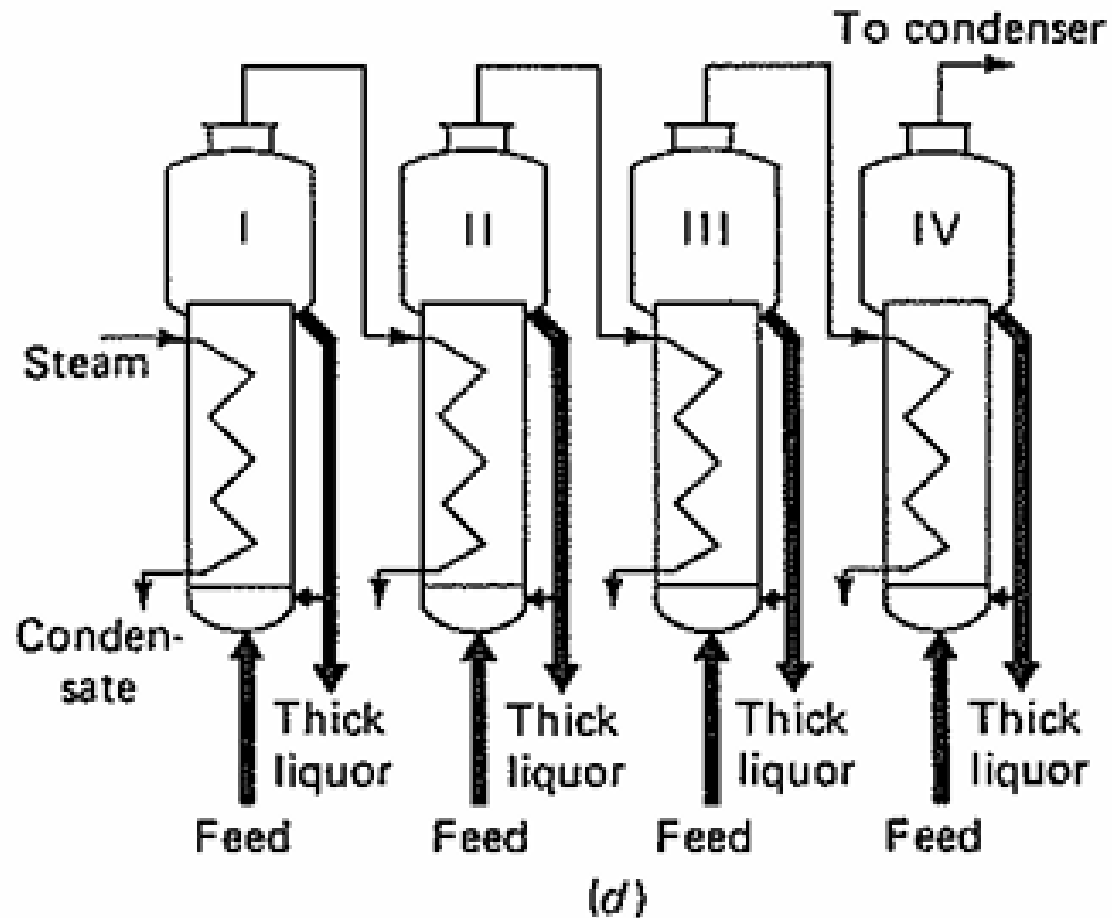
Methods of Feeding – Mixed Feed



(c)

- In *mixed feed* the dilute liquid enters an intermediate effect, flows in forward feed to the end of the series, and is then pumped back to the first effects for final concentration.
- This eliminates some of the pumps needed in backward feed and yet permits the final evaporation to be done at the highest temperature.

Methods of Feeding – Parallel Feed



- In crystallizing evaporators, where a slurry of crystals and mother liquor is withdrawn, feed may be admitted directly to each effect to give what is called *parallel feed*.
- In parallel feed there is no transfer of liquid from one effect to another.

Capacity and Economy of Multiple-effect Evaporators

- The increase in economy through the use of multiple-effect evaporation is obtained at the cost of reduced capacity.
- It might be thought that by providing several times as much heating surface the evaporating capacity would be increased, but this is not the case.
- The total capacity of a multiple-effect evaporator is usually no greater than that of a single-effect evaporator having a heating surface equal to one of the effects and operating under the same terminal conditions, and, when there is an appreciable boiling-point elevation, is often considerably smaller.
- When the boiling-point elevation is negligible, the effective overall ΔT equals the sum of the ΔT 's in each effect, and the amount of water evaporated *per unit area of surface* in an N-effect multiple-effect evaporator is approximately $(1/N)$ th that in the single effect.

Capacity and Economy of Multiple-effect Evaporators

- If the heating load and the heat of dilution are neglected, the capacity of an evaporator is directly proportional to the rate of heat transfer. The heat transferred in the three effects

$$q_1 = U_1 A_1 \Delta T_1 \quad q_2 = U_2 A_2 \Delta T_2 \quad q_3 = U_3 A_3 \Delta T_3$$

- The total capacity is proportional to the total rate of heat transfer q_T found by adding these equations.

$$q_T = q_1 + q_2 + q_3 = U_1 A_1 \Delta T_1 + U_2 A_2 \Delta T_2 + U_3 A_3 \Delta T_3$$

- Assume that the surface area is A square meters in each effect and that the overall coefficient U is also the same in each effect.

$$q_T = UA(\Delta T_1 + \Delta T_2 + \Delta T_3) = UA \Delta T$$

Capacity and Economy of Multiple-effect Evaporators

- Where ΔT is the total temperature drop between the steam in the first effect and the vapor in the last effect.
- Suppose now that a single-effect evaporator with a surface area A is operating with the same total temperature drop.
- If the overall coefficient is the same as in each effect of the triple-effect evaporator, the rate of heat transfer in the single effect is

$$q_T = UA \Delta T$$

- This is exactly the same equation as for the multiple-effect evaporator.
- No matter how many effects are used, provided the overall coefficients are the same, the capacity will be no greater than that of a single effect having an area equal to that of each effect in the multiple unit.

Capacity and Economy of Multiple-effect Evaporators

- In a single-effect unit producing 50 percent NaOH, for example, the overall coefficient U for this viscous liquid would be small.
- In a triple-effect unit, the coefficient in the final effect would be the same as that in the single effect, but in the other effects, where the NaOH concentration is much lower than 50 percent, the coefficients would be greater.
- Thus the average coefficient for the triple-effect evaporator would be greater than that for the single effect.
- In some cases this overshadows the effect of boiling-point elevation, and the capacity of a multiple-effect unit is actually greater than that of a single effect.